MODELS & EVALUATION

§3 LLM SIDE

Representation alignment · Temporal instruction & reasoning · Workflow policy learning

- ChatTS
- Time-LLM
- TimeRLM

§7 VALIDATION & EVALUATION

Execution checks · Information-cutoff checks · Task metrics · Downstream utility

- TimeSeriesGym
- Dr-CiK
- TraceBench
- TSFLab

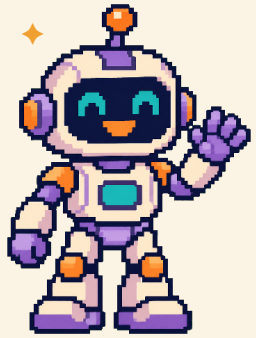
WHAT EVALUATION MUST SEPARATE

1. process validity
2. temporal validity
3. task performance
4. decision utility

Benchmarks expose the information, actions and output requirements the deployed workflow will meet (§7).

The Time Series Agent loop: how the parts work together

- MEMORY
- PLANNING
- TOOLS
- TIME



LLM × HARNESS × TIME × EVALUATION

LLM §3 → GH §4 → GTS §5 → TSK §6 → EVAL §7

↺ feedback: revise · or terminate

IN	Time series & context – observations · events · task request
OUT	Forecast · diagnosis · decision – accepted output

HARNESS: GENERAL → SPECIFIC

§4 GENERAL HARNESS

Interface & Interaction · Memory & Context · Tools & Execution · Planning & Orchestration · Optimization & Evolution

- TimeClaw-Harness
- AION
- TS-Agent
- TradingAgents

§5 GENERAL TIME SERIES HARNESS

Temporal Data Management · Temporal Processing · Temporal Context & Memory · Temporal Validation

- TimeSage-EV
- OpenForecaster
- TSQAgent
- TS-Reasoner

§6 TASK–SPECIFIC HARNESS

Forecasting & Prediction · Data Augmentation & Synthesis · Anomaly Detection & Diagnosis · Decision Support

- FLAIRR-TS
- MetaCaster
- Argos
- FinCon

EACH WORK MAY CONTRIBUTE TO SEVERAL LAYERS

Layer = scope implied by a module's inputs, state and assumptions.
Modules are tested from the most specific layer to the most general (Table 2).